

Exploring the social life of urban spaces through AI

Arianna Salazar-Miranda, Zhuangyuan Fan, Michael Baick, Keith N. Hampton, Fabio Duarte, Becky P. Y. Loo, Edward Glaeser, and Carlo Ratti, 2025

Computational Social Science with Images and Audio - ETH Zurich, Fall Term 2025, Course No. 851-0762-00L
Lecturers: Elliott Ash & Philine Widmer

Vanessa Costalonga

Lena Libon

- What is the research question?
- How is it answered (data, methods)?
- (How) Does the approach relate to the material covered in class so far?
- What are the main results? Are they convincing? Why (not)? What do you think are the broader implications of this work (in research or society/politics), and why? For example, how could the used methods inspire future research? Are there important policy implications? Do you have any ethical concerns about the research?
- On your last slide, prepare 1-2 questions to discuss in class.

The Social Life of Small Urban Spaces (Whyte; 1980)

Observational study of plazas and sidewalks



Methods:

- Time-lapse film and manual coding
- Documented: movement, lingering, group formation
- Focus on micro-behaviors shaping social life

Results: Small features have big impacts

- Seating, food vendors, visibility, trees, water, other people attract people
- Blank walls, hidden plazas fail



Bryant Park

Images derived from Salazar-Miranda et al. 2025, page 3 (top) and taken by Jean-Christophe BENOIST (bottom)



1980

30 years
later



2010

Whyte' findings:

- Documented movement, lingering, group formation
- Showed importance of small, everyday interactions
- Proved that public spaces shape social life
- Relied on manual, labor-intensive coding

New Questions:

- Are sidewalks still social, or just for movement?
- Have walking speeds increased?
- Do people still linger and meet in groups?
- What do changes reveal about urban life today?

Relevance:

Social life on sidewalks may be changing

Time pressure and technologies reshape behavior

Need evidence to guide future urban design

Advantage of AI & Computer Vision:

Automates analysis

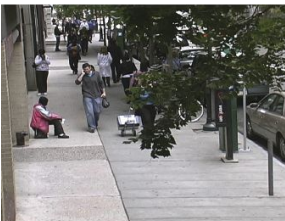
Scales across decades

Reduces subjectivity

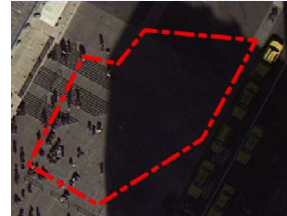
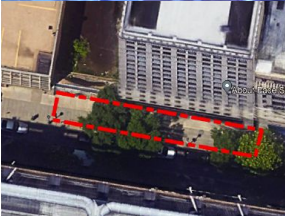
Do city sidewalks serve primarily for pedestrian mobility, or do they also serve as social spaces to connect people? And has their function changed over time?

Analysed Locations

2010s



1980s



Chestnut Street
Philadelphia

Downtown Crossing
Boston

Bryant Park
New York City

Metropolitan Museum of Art
New York City

Dataset

1980s



Archival footage

*Super 8 films captured by
Whyte, digitized*

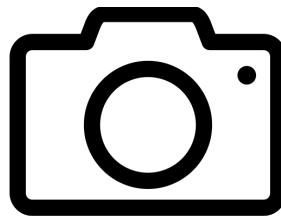
Taken from rooftops and
windows

Adjusted frame-rates to
match 10s

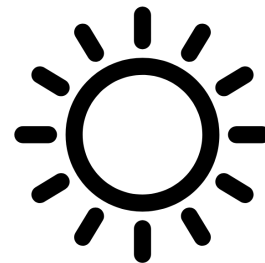
2010s



Lower altitude



Unobstructed and
stationary camera

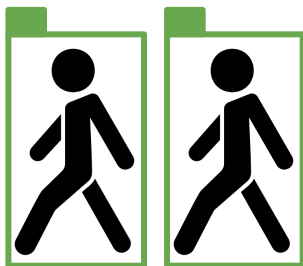


Filmed at midday,
Except Chestnut
(morning)

638

total footage
minutes

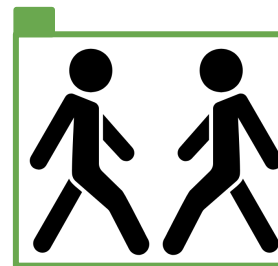
Features



Pedestrian Tracking
and Classification

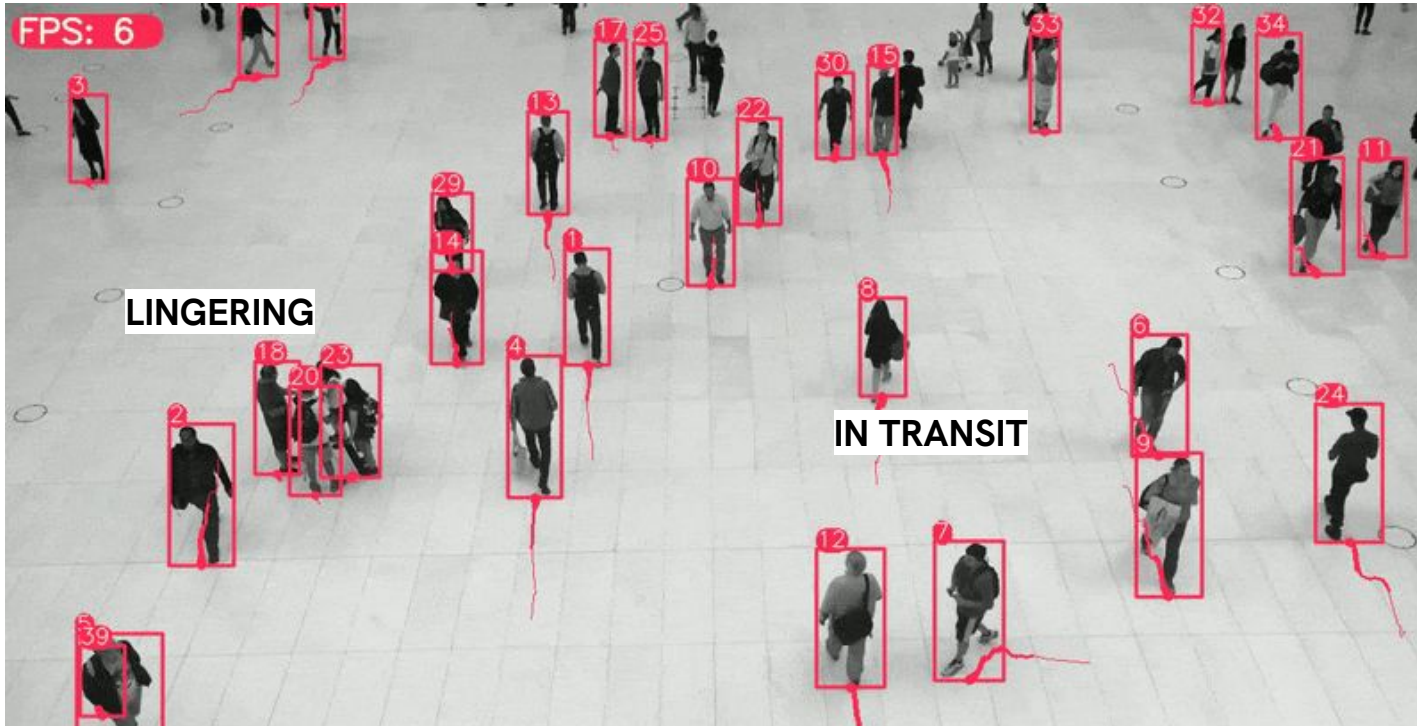


Group Detection



Identification of
Encounters

Pedestrian Tracking



Performance
Validation (YOLO):

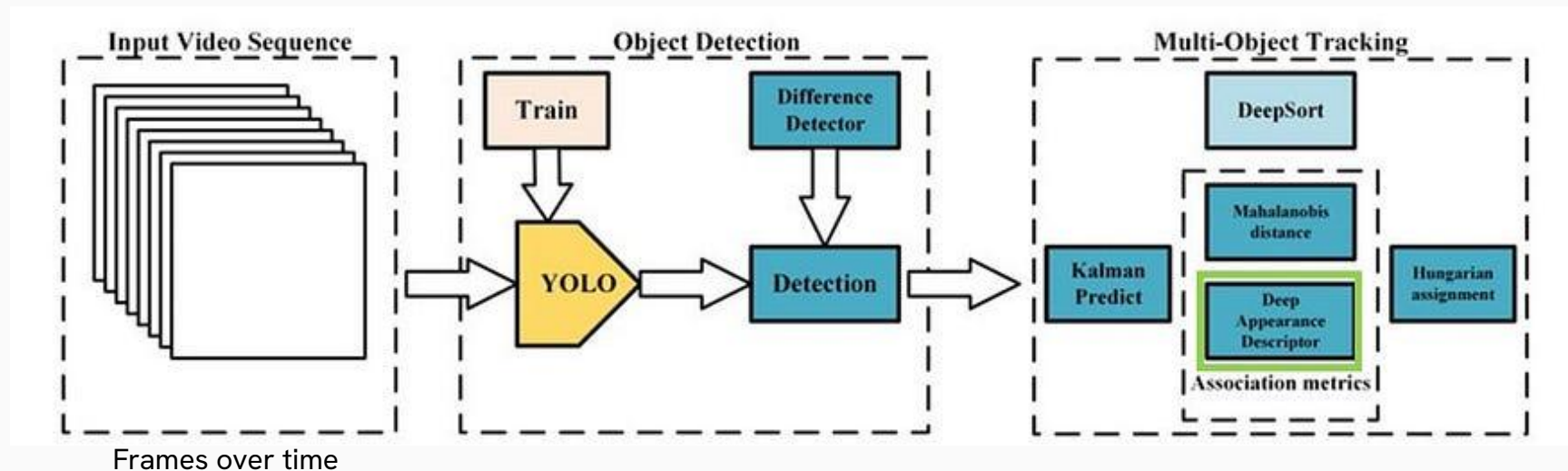
- Precision
- Recall
- Mean average precision
- Precise for 2010s, less for 1980s
- Manually labelled images

Diagrams made with pictogram from Darwin Mulyan for Noun Project - thenounproject.com

Gif from <https://muhammadrizwanmunawar.medium.com/as-one-reduced-projects-development-of-computer-vision-ae83f5cb0903>

DeepSORT (Deep Simple Online and Realtime Tracking)

- Multi-object tracking algorithm → Each object gets a unique, persistent ID across frames



- **Object detection:** Use YOLO to find objects in each video frame
- **Kalman filter:** Predict where each tracked object will move in the next frame (based on velocity, direction, ...)
- **Data association:** Match new detections with existing tracks
 - Mahalanobis distance: Motion similarity (position and velocity)
 - Deep appearance descriptor:
- **Hungarian algorithm:** Solve the assignment problem optimally (who is who)

Group Detection

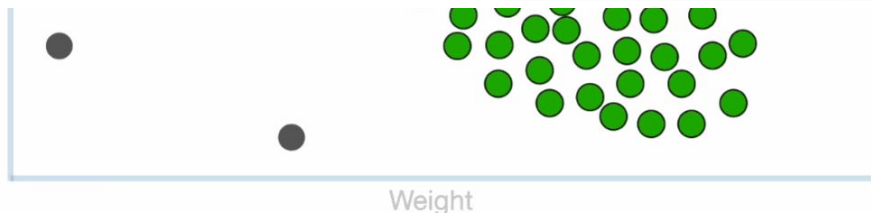


...by identifying two different, but relatively



Validation:

- Manual analysis of 80 frames (3000 pedestrians) from each location from 1980 and 2010
- Accuracy of 87.91%



Identification of Encounters

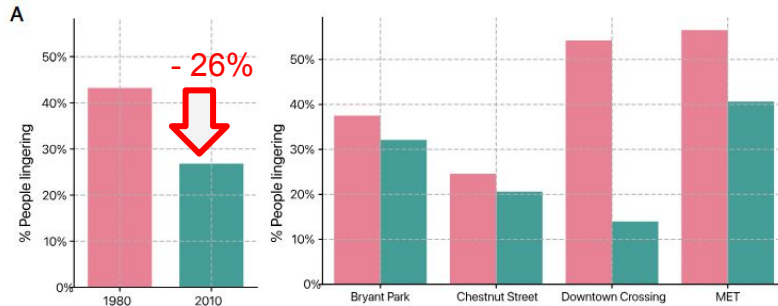


Fig. S4. Example of detected encounters. Images from left to right show frames captured every 1.5 seconds. The entire sequence illustrates an encounter lasting around 6.5 seconds as two pedestrians approach and meet.

Results

Result 1.1: People linger less in public spaces

People lingering per frame (< 0.5 m/s)



→ visible across all locations analyzed

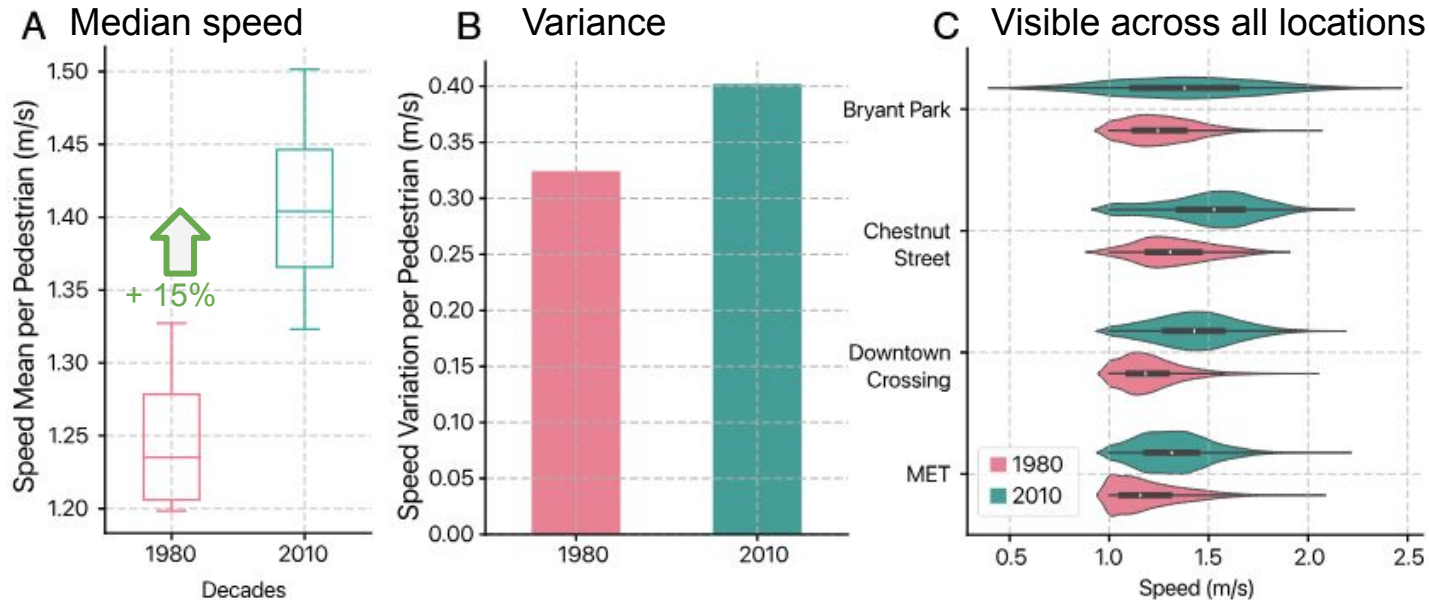
Singles lingering per frame (< 0.5 m/s)



→ similar decline to that seen for all groups
→ decline in lingering holds even when group composition is removed from the equation

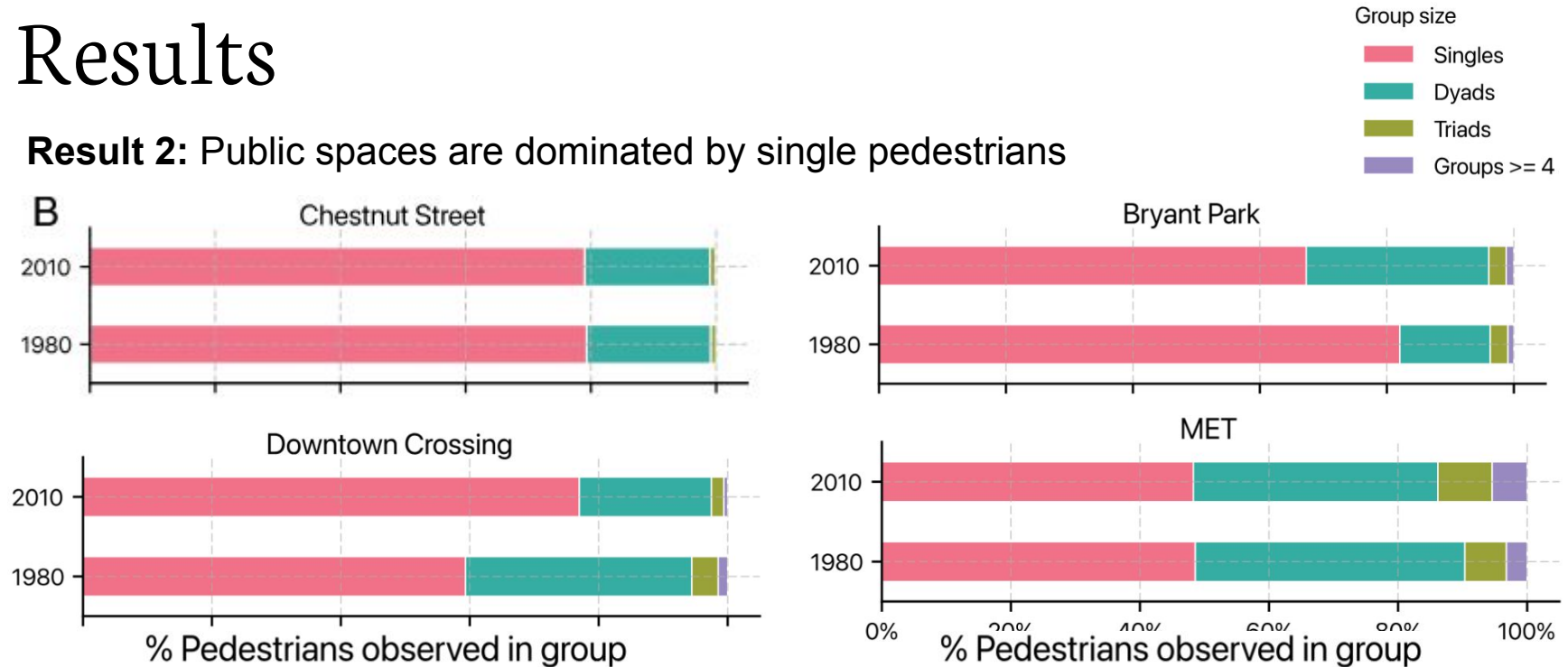
Results

Result 1.2: People walk through the sites faster



Results

Result 2: Public spaces are dominated by single pedestrians

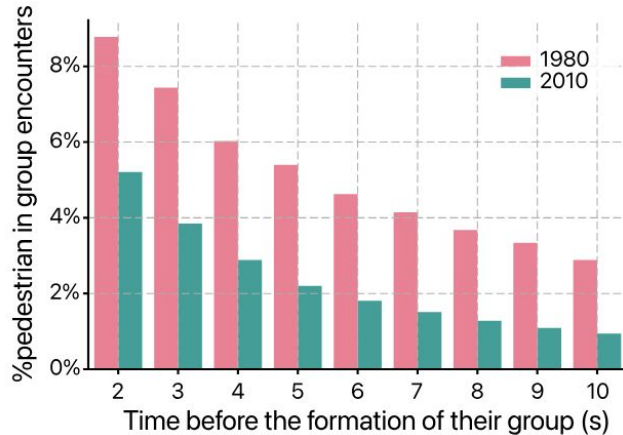


- Many tourists in groups at Metropolitan Museum of Art (MET)
- Chestnut Street primarily pedestrian transit corridor between destinations

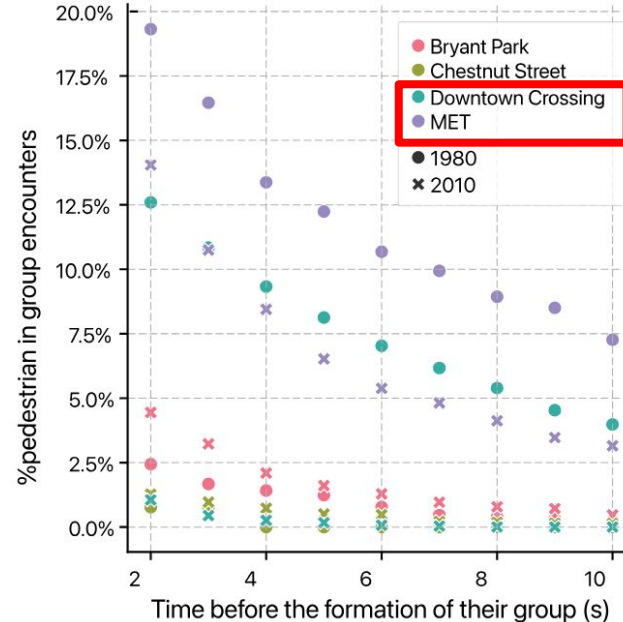
Results

Result 3: Decline in Group Encounters

Groups: Preplanned meetings or spontaneous encounters → can't distinguish



Share of people that form at least one group s seconds after they enter the location



Broad changes in how places are used

Summary



Single individuals
Less groups forming



Less time



Faster pace

Analysis of results (why?)



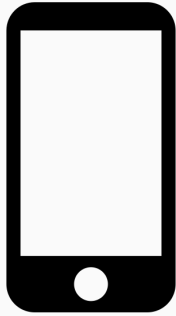
Higher wages might be associated with increased walking speeds and reduced social interactions



Exception!
The MET = high levels of lingering despite high-income neighborhood

Compelling
urban
amenities
=
Slower social
behaviour

Analysis of results (why?)



Smartphones changed
people's behavior?

Tested CV Model

Low phone usage



Decrease in homelessness?

Tested CV Model

No evidence in 1980s



More coffee shops/indoor
venues that keep people
indoors?

No data available

Limitations

Number of locations surveyed is limited:
Intentional to compare with original 1980s study

Some factors (gender, income etc), remain a
challenge to identify

Does not capture quality of interactions

Speed threshold is too short could be
excluding slower individuals (i.e. elderly)

Privacy: Public footage without consent - no
ethical reflection on paper

Future Research

Look at different areas, including different seasons
and time of day

Focus on different population groups (more women
walking alone could be a sign of safer streets?)

Use larger datasets and more nuanced categories of
social interactions, potentially incorporating other
data sources (Bluetooth, GPS)

Study parameters with diverse group composition

Investigate anonymization methods/ consider
ethics of public data collection

Question 1

If you repeated the analysis for 2023–2025, post-pandemic, what hypotheses would you test about lingering, speed, and group encounters?
What additional data sources would you integrate?

Question 2

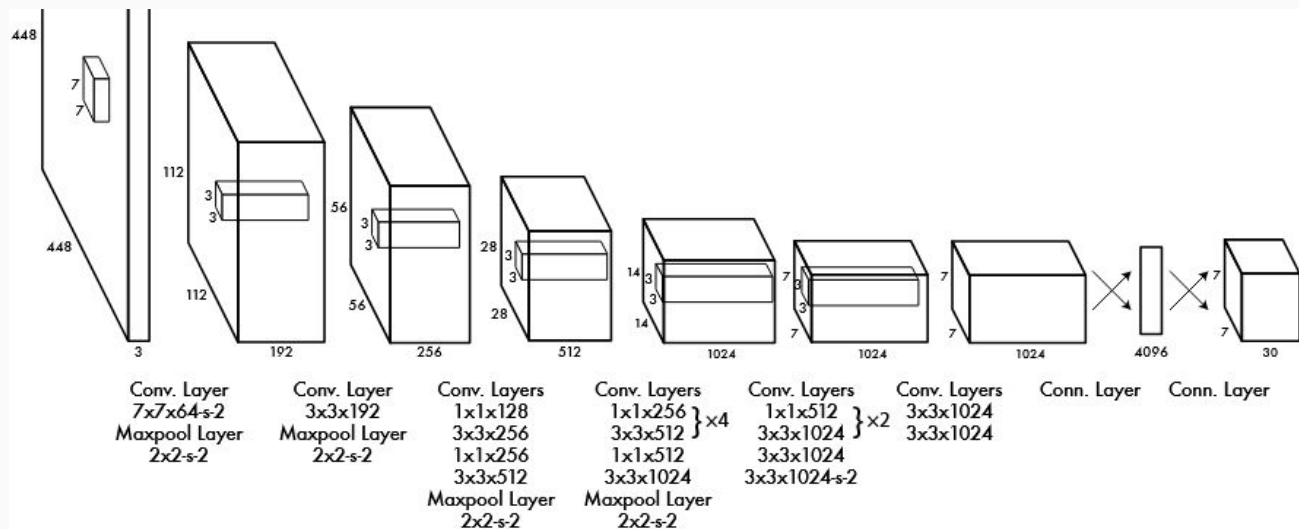
Could changes in retail composition, public transportation, policing, or urban design elements such as seating and edges explain the trends better than income? How would you test that?

Thank you!

Additional Slides

YOLO (You Only Look Once)

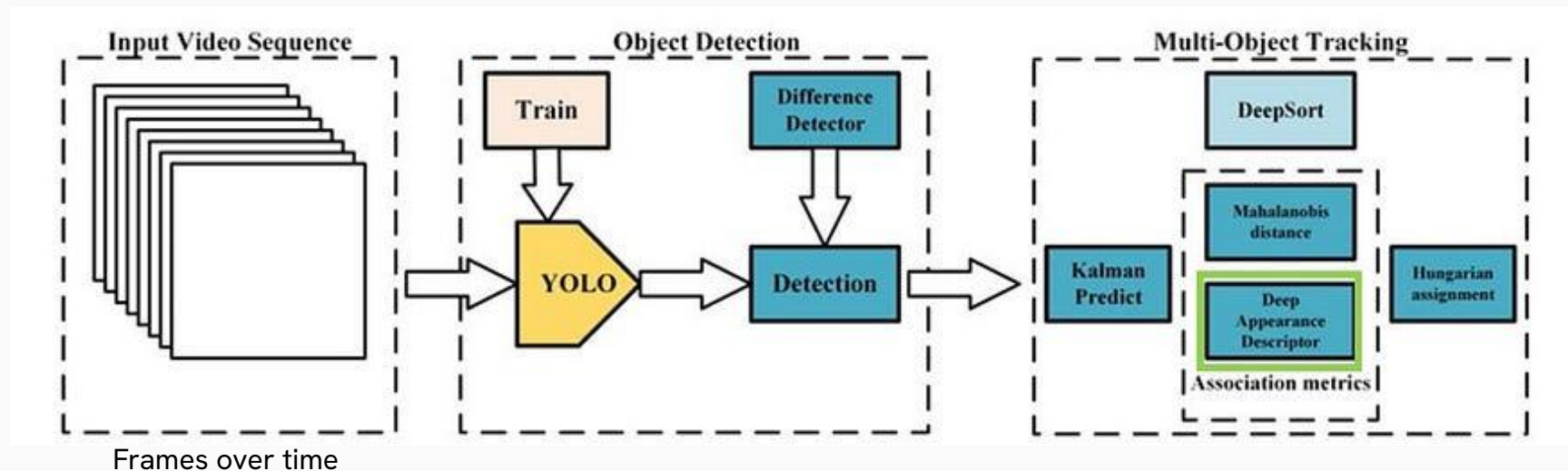
- CNN-based object detection model
- Requires **only one forward pass** to full image, instead of scanning an image region by region



- 1) Resize img to 448x448x3 (RGB)
- 2) Feature extraction (Convolution + MaxPool) and Fully connected layers → Final prediction tensor: 7x7x(B*5 + C)
 - B: Number of bounding boxes per cell (e.g. 2 in YOLOv1)
 - 5: 4 box coordinates (x, y, w, h) + confidence
 - C: Number of classes (20 in PASCAL VOC) for object class probabilities

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